FIELD OBSERVATION REPORT

Waaree Energies Ltd.

Solar Photovoltaic Module Manufacturing Plant

Chikhli, Gujarat, India

Prepared by: Aarav Jain

Date of Visit: 23rd – 25th March 2026

WAAREE®
One with the Sun

Date: - 07th May, 2026

To Whomsoever It May Concern

I am pleased to recommend **Aarav A Jain**, Grade 11 of Step-by-Step School, Noida, who undertook an in-depth academic engagement involving a field study of solar photovoltaic manufacturing at Waaree Energies Ltd.

During the engagement from 23rd to 25th March 2026, Aarav gained exposure to industrial processes, quality control, and organizational practices, including ISO standards and 6S methodology. He learned semiconductor basics and the end-to-end production flow—from cell fabrication and module assembly to quality testing and high-volume manufacturing. He also showed strong interest in advanced solar technologies such as PERC and TOPCon. Throughout, he demonstrated intellectual curiosity and the ability to connect classroom concepts with real-world industrial applications.

His ability to grasp knowledge of integrated principles of physics and materials science, was commendable. Aarav's ability to synthesize complex, interdisciplinary concepts into a structured and analytical report was especially impressive. His engagement was clearly guided by a genuine commitment to environmental sustainability, which led him to explore solar energy as a practical and scalable solution to environmental problems.

Beyond his technical skills, Aarav showed maturity and professionalism throughout the engagement. He remained disciplined and eager to learn, asked thoughtful questions, contributed to cross-department discussions, and clearly communicated his observations and conclusions.

Aarav is a motivated and intellectually engaged person who brings both curiosity and rigor to his work. I believe this industrial exposure will be helpful in shaping his future career goals. I am confident that he will excel in academically challenging environments and contribute meaningfully to any institution he joins.

Waaree Energies Limited

WAAREE ENERGIES LIMITED

Vishal Upadhyay
Vice President- Human Resource

ABSTRACT

This report presents a comprehensive field observation of the Waaree Energies Ltd. solar photovoltaic (PV) module manufacturing facility located at Chikhli, Gujarat, India. The visit, conducted over three days, provided access to the module-level (MLB) production line, the cell-level (CLB) manufacturing process, quality control and testing laboratories, the Human Resources department, and the maintenance division. Waaree Energies, founded in 1989 and pivoting to solar energy in 2007, has grown to become India's largest solar module manufacturer with an installed capacity of 23.75 GW as of 2025. This report examines the complete photovoltaic value chain — from silicon semiconductor physics and wafer fabrication, through the PERC and TOPCon cell manufacturing processes, to module assembly, lamination, quality testing, and dispatch — supported by detailed scientific theory in semiconductor physics, photovoltaic operation, materials science, and thin-film deposition. Concurrently, the report analyses Waaree's business structure, growth trajectory, quality management systems (ISO, 6S, SOP), and human capital management. The observation reveals a highly automated, vertically integrated manufacturing ecosystem operating at industrial scale with a clear strategic orientation toward green hydrogen and end-to-end clean energy solutions.

Keywords: *photovoltaic; PERC; TOPCon; solar module manufacturing; silicon semiconductor; EVA encapsulation; IV curve; sun simulator; 6S; ISO 9001; Waaree Energies; green hydrogen.*

Founded	1989 (measuring instruments); Solar since 2007
Facility Visited	Chikhli Manufacturing Campus, Gujarat
Plant Capacity	5.4 GW cell manufacturing capacity (Chikhli)
Total Company Capacity	23.75 GW (as of 2025)
Products Manufactured	PERC & TOPCon PV Modules, Electrolysers, Transformers
Report Type	Formal Scientific & Business Field Observation
Subject Areas	Photovoltaic Technology / Materials Science / Manufacturing

1. Introduction and Company Background

Waaree Energies Ltd. is India's largest solar photovoltaic (PV) module manufacturer by installed capacity, headquartered in Mumbai with its primary manufacturing campus in Chikhli, Gujarat. The company's trajectory from a precision instrument manufacturer to a fully integrated clean energy conglomerate represents one of the most significant industrial pivots in India's renewable energy sector over the past two decades.

This three-day field visit to the Chikhli facility provided an immersive, multi-departmental observation of the entire solar module manufacturing ecosystem — from semiconductor physics and cell fabrication through module assembly, quality certification, human capital management, and corporate strategy. The observations documented in this report reflect activities across the Module-Level production Line (MLB), the Cell-Level production Line (CLB), the testing laboratory, the Human Resources department, and the maintenance division, as outlined in the visit schedule.

1.1 Company Timeline and Growth Trajectory

Waaree's evolution from instrument manufacturer to solar energy giant is best understood through its strategic milestones:

Year	Milestone	Significance
1989	Founded; manufacturing measuring instruments and pressure gauges	Industrial base; precision engineering capabilities developed
2007	Entry into solar energy; 30 MW initial capacity	Strategic pivot to renewable energy; leveraged existing manufacturing expertise
2011	Engineering, Procurement & Commissioning (EPC) services launched	Vertical integration; end-to-end project capability
2014	Joint Venture with NEEPCO (North Eastern Electric Power Corporation)	Government sector partnerships; North-East India expansion
2018	Capacity reaches 1 GW	Scale inflection point; industrial-grade manufacturer status
2021–2023	Capacity scaled to 12 GW	Major capacity expansion; response to India's PLI scheme for solar
2025	23.75 GW total capacity; green hydrogen ventures initiated	Market leadership; diversification into electrolyser manufacturing

Table 1.1 — Waaree Energies chronological growth milestones.

1.2 Corporate Scale and Financial Overview

As of 2025, Waaree Energies commands a market capitalisation of approximately ₹25,000 crore and annual revenues of approximately ₹7,500 crore. The company operates nine manufacturing facilities across India, producing modules, electrolysers, silicon wafers, transformers, and associated components. Its workforce exceeds 10,000 employees, with approximately 80% in direct operational roles.

The strategic positioning of Waaree as an 'end-to-end solutions ecosystem' — encompassing module manufacturing, EPC contracting, green hydrogen production via electrolysis, and power trading — reflects a deliberate vertical integration strategy designed to capture value across the entire clean energy value chain rather than competing solely on module commodity pricing.

1.3 The Chikhli Facility

The Chikhli campus in Gujarat is Waaree's flagship manufacturing site, spanning 150 acres and housing 5.4 GW of cell manufacturing capacity — the largest single-site PV cell manufacturing facility in India. The facility integrates the complete value chain from silicon wafer processing through to finished module testing and dispatch. The scale and automation level observed during the visit were consistent with Tier-1 global PV manufacturers.

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2. Scientific Foundations: Photovoltaic Physics

Before examining the manufacturing processes observed during the visit, it is essential to establish the scientific principles that underpin the operation of a solar photovoltaic cell. These principles — drawn from semiconductor physics, quantum mechanics, and materials science — directly govern every manufacturing decision, process parameter, and quality specification observed at the plant.

2.1 Silicon as a Semiconductor

Silicon (Si, atomic number 14) is the foundational material of virtually all commercial solar cells, occupying approximately 95% of the global PV market. It belongs to Group 14 of the periodic table and forms a diamond cubic crystal structure in its pure (intrinsic) form, with each silicon atom covalently bonded to four neighbours in a tetrahedral arrangement.

As a semiconductor, silicon possesses an electrical conductivity intermediate between conductors and insulators, governed by its band structure. In quantum mechanical terms, the electrons in a solid crystal occupy energy bands separated by forbidden energy gaps (band gaps). The valence band is the highest energy band fully occupied by electrons at 0 K; the conduction band is the next higher band, normally empty. The band gap of silicon is 1.12 eV — a critical property that determines its interaction with the solar spectrum.

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E_g(Si) = 1.12 eV   (at 300 K)
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Photons from the sun with energy greater than or equal to the band gap (E_photon = hf ≥ E_g) can excite electrons from the valence band to the conduction band, generating an electron-hole pair. Photons with energy below 1.12 eV (infrared radiation, wavelength > ~1,100 nm) pass through silicon without absorption. Photons with energy significantly above 1.12 eV are absorbed but the excess energy is lost as heat (thermalisation) — a fundamental thermodynamic limit of single-junction solar cells.

> **Note:** *The Shockley-Queisser limit predicts a maximum theoretical efficiency of approximately 33% for a single-junction silicon solar cell under standard solar spectrum (AM1.5G). Current commercial PERC cells achieve ~22–24%; TOPCon cells reach ~24–26%. Waaree's observed plant efficiency of ~25% is therefore consistent with state-of-the-art commercial production.*

2.2 Doping: Creating Extrinsic Semiconductors

Pure (intrinsic) silicon has very few free charge carriers at room temperature and is therefore a poor conductor. To create a functional solar cell, controlled impurities (dopants) are intentionally introduced into the silicon crystal lattice — a process called doping — creating extrinsic semiconductors with greatly enhanced conductivity and, critically, with a defined majority charge carrier type.

2.2.1 P-Type Silicon (for PERC cells)

Boron (B, Group 13) is the standard p-type dopant. Boron has three valence electrons; when it substitutes for a silicon atom (four valence electrons) in the crystal lattice, it creates a 'missing electron' or positive hole. These holes are the majority charge carriers in p-type silicon. At Waaree, boron trichloride (BCl_3) gas is used as the boron source in the diffusion furnace. The chemical reaction during diffusion is:

$$4BCl_3 + 3O_2 \rightarrow 2B_2O_3 + 6Cl_2 \quad \text{(oxidation of boron source)}$$

$$2B_2O_3 + 3Si \rightarrow 4B + 3SiO_2 \quad \text{(reduction; boron diffuses into Si lattice)}$$

2.2.2 N-Type Silicon (for TOPCon cells)

Phosphorus (P, Group 15) is the standard n-type dopant. With five valence electrons, a phosphorus atom substituting for silicon in the lattice donates one extra electron to the conduction band, making electrons the majority carriers. Phosphorus oxychloride ($POCl_3$) is used as the phosphorus diffusion source:

$$4POCl_3 + 3O_2 \rightarrow 2P_2O_5 + 6Cl_2$$

$$2P_2O_5 + 5Si \rightarrow 4P + 5SiO_2$$

Note: *The choice between PERC (p-type base) and TOPCon (n-type base) fundamentally determines the cell architecture and manufacturing process sequence. N-type silicon has lower bulk recombination rates (longer minority carrier lifetimes) than p-type, conferring a fundamental efficiency advantage to TOPCon — which is why TOPCon is increasingly the industry's preferred technology for high-efficiency applications.*

2.3 The p-n Junction and Photovoltaic Effect

A solar cell operates by creating a p-n junction — an interface between a p-type and an n-type semiconductor region within the same silicon crystal. At this junction, electrons from the n-type region diffuse across to combine with holes in the p-type region, and vice versa, creating a depletion region devoid of free charge carriers. This charge separation creates a built-in electric field directed from the n-type to the p-type region, with a built-in potential of approximately 0.5–0.7 V for silicon.

When a photon with energy ≥ E_g is absorbed near the junction and generates an electron-hole pair, the built-in electric field separates the carriers before they can recombine: electrons are swept to the n-type side and holes to the p-type side. This charge separation drives a photocurrent (I_ph) through an external circuit — this is the photovoltaic effect.

The electrical behaviour of an illuminated solar cell is described by the ideal diode equation with a photocurrent source:

$$I = I_{ph} - I_0 \left[\exp(qV/nkT) - 1\right]$$

where I is the cell current, I_ph is the photogenerated current, I_0 is the dark saturation current, q is the elementary charge (1.602×10^{-19} C), V is the terminal voltage, n is the ideality factor (1–2), k is Boltzmann's constant (1.381×10^{-23} J/K), and T is temperature in Kelvin. This equation defines the characteristic IV curve of a solar cell, which is the fundamental measurement used in the plant's testing laboratory.

2.4 Key Cell Parameters: The IV Curve

The current-voltage (IV) characteristic curve encodes all the essential performance parameters of a solar cell. In the testing laboratory at Waaree, a flash xenon lamp solar simulator recreates the standard AM1.5G solar spectrum at an irradiance of 1,000 W/m² to measure this curve. The key parameters extracted from the IV curve are:

Parameter	Symbol	Definition	Typical Value (PERC)
Short-circuit current	I_{sc}	Current when V=0; proportional to photon flux	~10–12 A (full cell)
Open-circuit voltage	V_{oc}	Voltage when I=0; logarithmically dependent on I_{ph}/I_0	~0.67–0.70 V
Maximum power point current	I_{mp}	Current at maximum power output	~9.5–11 A
Maximum power point voltage	V_{mp}	Voltage at maximum power output	~0.59–0.62 V
Maximum power	P_{max}	$P_{mp} = I_{mp} \times V_{mp}$; rated power of cell	~7.6 W (182×182 mm)
Fill Factor	FF	$FF = P_{mp} / (V_{oc} \times I_{sc})$; measure of curve squareness	~80–83%
Efficiency	η	$\eta = P_{mp} / (G \times A)$; G = irradiance, A = cell area	~22–25%

Table 2.1 — Key IV curve parameters and their physical significance.

At Waaree, the 182×182 mm (M10 format) full-size cell generates approximately 7.6 W, while the half-cut 182×91 mm variant produces approximately 3.8 W. A standard module containing 144 half-cut cells therefore has a rated output of approximately 144 × 3.8 = 547 W — consistent with Waaree's G12 and M10 module product lines.

3. Day 1 Observations — MLB Production Line & Safety Induction

3.1 Safety Induction

The first activity on Day 1 was a mandatory safety induction conducted by the Health, Safety and Environment (HSE) team. This is standard practice at ISO-certified manufacturing facilities and is a prerequisite for any individual accessing the production floor. The induction covered:

- Personal Protective Equipment (PPE) requirements: safety glasses, anti-static footwear, and high-visibility vests mandatory in production zones.
- Emergency evacuation routes and assembly points for the 150-acre campus.
- Electrostatic discharge (ESD) protocols: silicon wafers and assembled cells are extremely sensitive to electrostatic discharge, which can create micro-defects in the p-n junction invisible to the naked eye but detectable by electroluminescence testing. All personnel and equipment in the cell line are grounded.
- Chemical hazard awareness: the cell manufacturing line uses hazardous process gases including $POCl_3$, BCl_3, SiH_4 (silane), NH_3, and HF — all managed under MSDS protocols.
- Compressed air and pneumatic system safety: the robotic handling systems use pneumatic suction cups, and compressed air lines operate at pressures requiring specific isolation procedures.

3.2 Introduction to Waaree — Corporate Presentation

Following the safety induction, a corporate presentation was delivered by a senior company representative. This session covered Waaree's history, strategic positioning, product portfolio, and future direction. The key themes communicated were:

- Vertical integration: Waaree's differentiation from competitors lies in its ability to manufacture across the entire value chain — wafers, cells, modules, transformers, and electrolysers — from a single integrated campus.
- End-to-end solutions: the company positions itself not as a component supplier but as a complete energy solutions provider, offering EPC (Engineering, Procurement and Commissioning) services and long-term O&M (Operations and Maintenance) contracts.
- Green hydrogen: Waaree's investment in electrolyser manufacturing represents a forward-looking bet on hydrogen as a clean fuel, utilising the company's existing competence in precision electrical manufacturing.
- Domestic manufacturing imperative: the presentation emphasised India's Production-Linked Incentive (PLI) scheme for solar manufacturing as a policy tailwind, and Waaree's alignment with the Government of India's goal of 500 GW renewable energy capacity by 2030.

3.3 MLB (Module-Level Production Line) — Process Overview

The Module-Level production Line (MLB) is the downstream assembly stage where individual solar cells are interconnected and encapsulated into finished PV modules. The MLB at Chikhli is a highly automated line utilising robotic handling, automated optical inspection, and computer-controlled process stages. The following sections describe each stage in detail as observed during the line visit.

3.3.1 Cell Receiving and Preparation

Solar cells arrive at the MLB assembly area pre-manufactured from the CLB (Cell-Level production Line) or sourced from external cell suppliers. Cells are delivered in protective packaging boxes, with individual cells separated by interleaved butter paper (smooth paper sheets) to prevent abrasion between cell surfaces. Cells are fed into the MLB through a hopper — a gravity-fed input system — from which they enter an alignment station.

A critical preparatory step immediately follows: each 182×182 mm full-size cell is laser-cut in half, producing two 182×91 mm half-cells. The rationale for this step is fundamental to module electrical design:

- Half-cell modules have half the current per cell string ($I \propto$ cell width) compared to full cells, which reduces resistive (ohmic) power losses in the busbars by a factor of four ($P_{loss} = I^2R$). Since the cells are connected in parallel strings, halving I halves I^2 and quarters losses.
- Reduced cell-level mechanical stress from thermal cycling — smaller cells experience lower differential thermal expansion forces.
- Better shading tolerance — partial shading of a half-cell module affects a smaller fraction of the total string current.

The laser cutting process uses water as a cooling medium during the cut, followed by thermal treatment to evaporate residual moisture. Crucially, one of every two half-cell pieces is flipped (rotated 180°) before insertion into the module — this ensures uniformity of current flow direction across the module and correct orientation of the p and n terminals relative to the interconnection ribbons.

A CCD (Charge-Coupled Device) camera performs automated optical inspection (AOI) at this stage, detecting any micro-cracks, edge defects, or surface contamination introduced during laser cutting.

3.3.2 Stringer — Cell Interconnection

The stringer is the stage at which individual half-cells are electrically interconnected using copper ribbons to form cell strings — linear chains of cells connected in series. The ribbons are copper wires coated with a lead-tin (Pb-Sn) solder alloy, flattened to a cross-section of approximately 0.4 mm thickness to reduce optical shading while maintaining adequate electrical conductivity.

Before soldering, the ribbons are passed through a flux bath. Flux is a transparent chemical compound (typically rosin or organic acid based) that serves two functions: it chemically removes the thin oxide layer from both the copper ribbon surface and the silver busbars on the cell, ensuring a metallurgically clean solder joint, and it reduces surface tension of the molten solder to improve wetting. The flux-coated ribbons are aligned precisely with the cell's busbars (the silver conducting lines on the cell surface) and passed through a belt furnace — a continuous heating unit — where infrared lamps heat the assembly to the solder melting point (~180–200°C), creating a permanent metallurgical bond.

The bus bar configuration observed at the plant includes three types: I-type (straight), Z-type (offset), and U-type (return), depending on their position and function within the module circuit. The strings produced at the stringer stage consist of 12 half-cells each, which will subsequently be assembled into the final module layout.

> $P_{loss(ribbon)} = I^2 \times R \quad \text{where } R = \rho L/A \quad \text{(Ohm's law for resistive losses)}$

Automatic current testing is performed on each completed string immediately after soldering to verify the current flow path and identify any open-circuit or high-resistance solder joints before they proceed to module assembly.

3.3.3 Module Lay-Up — Sandwich Assembly

Completed strings are assembled into the full module 'sandwich' structure on the lay-up table, where each layer is added in precise sequence. The module structure, from bottom to top as assembled, consists of:

Layer (bottom to top)	Material	Function
Back glass / backsheet	Toughened glass or polymer backsheet	Structural support; moisture barrier; electrical insulation
Bottom EVA sheet	Ethylene Vinyl Acetate (EVA) polymer	Encapsulant; adheres to back glass; cushions cells
Cell strings	PERC or TOPCon half-cells with copper ribbons	Photovoltaic active layer; generates electricity
Top EVA sheet	Ethylene Vinyl Acetate (EVA) polymer	Encapsulant; adheres to front glass; seals cells
Front glass	Anti-reflective toughened glass (>95% transmittance)	Structural protection; maximum light transmission

Table 3.1 — PV module sandwich structure: layers, materials and functions.

Movement of all components during lay-up is performed by robotic arms equipped with pneumatic suction cups — vacuum-based grippers that hold glass and cell layers without contact pressure that could crack the brittle silicon cells. Conveyer belts transport assemblies between stations. The glass sheets are lifted individually using suction cups after being separated from stacked pallets.

EVA (ethylene vinyl acetate, $[\text{-CH}_2\text{-CH}_2\text{-}]_x[\text{-CH}_2\text{-CH(OOCCH}_3\text{)-}]_y$) is the encapsulant of choice due to its combination of optical transparency, flexibility, strong adhesion, UV stability, and moisture resistance. Once melted and cured in the laminator, EVA encases all cell interconnections and provides mechanical protection against vibration, thermal cycling, and moisture ingress — the primary degradation mechanisms over the module's 25-year design life.

The interconnecting ribbons between adjacent cell strings (called cross-ribbons or bus ribbons) are also installed at this stage, completing the electrical circuit of the full 12×12 half-cell (144 cell) module. Alignment of all layers is verified by automated optical checks before the sandwich proceeds to the laminator, and the assembled stack is held together with temporary tape to prevent layer displacement during transport.

3.3.4 Laminator

The laminator is the most energy-intensive stage in the MLB process and the most critical for long-term module reliability. The assembled sandwich is placed in the laminator — a dual-chamber press with a flexible membrane — where a precisely controlled thermal cycle bonds all layers into an integrated, hermetically sealed structure.

The lamination cycle proceeds in two stages:

- Stage 1 — Vacuum and initial heating (to ~120°C): the upper and lower chambers are evacuated to remove air from within the sandwich. Air trapped between layers would form bubbles in the cured EVA, creating optical losses and potential delamination sites. The vacuum ensures intimate contact between all interfaces.
- Stage 2 — Pressurised heating (to ~150°C): atmospheric pressure is restored to the upper chamber while the lower remains evacuated, creating a pressure differential that presses the membrane firmly against the top glass. At 150°C, the EVA polymer crosslinks (polymerises) and transitions from a white, opaque sheet to a transparent, elastic solid. The crosslinking reaction creates covalent bonds between EVA polymer chains, producing the cohesive, inseparable encapsulation layer.

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EVA crosslinking: peroxide initiator (organic peroxide) →
                  free radicals → C-C crosslinks
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After lamination, the module is cooled in a controlled manner — rapid cooling would introduce thermal stresses in the glass layers. The laminator is the highest power consumer on the line, running continuously at elevated temperatures. Its throughput directly determines the maximum module output rate of the entire MLB.

3.3.5 Framing and Junction Box Installation

Post-lamination, the module receives its aluminium frame and junction box — the two components that transform it from a laminated panel into a commercially deployable module.

Aluminium frame: an extruded aluminium alloy frame is fitted to the perimeter of the laminated panel using a silicone sealant (applied as a liquid and cured over approximately 4 hours). Aluminium is selected for its combination of low density (2.7 g/cm³), high corrosion resistance (due to native Al_2O_3 passivation layer), good thermal conductivity (facilitating heat dissipation from the module), and relatively low cost. The frame provides structural rigidity, protects the glass edges from chipping, and provides the mounting interface for field installation. Aluminium's electrical conductivity also ensures the frame can be grounded for safety.

Junction box: the junction box (J-box) is a weatherproof enclosure mounted on the rear surface of the module that houses the bypass diodes and the DC output terminals. Bypass diodes (typically three Schottky diodes for a 144-cell module) are connected anti-parallel across groups of cells. When a group of cells is shaded and would otherwise act as a load (consuming power from unshaded cells, causing local heating — the 'hot spot' effect), the bypass diode conducts and short-circuits the shaded string, protecting the cells from thermal

damage. The J-box is the point at which the module's electrical output cables (MC4 connectors) are terminated.

4. Day 2 Observations — Human Resources & Testing Laboratory

4.1 Human Resources Department

The Day 2 programme commenced with a presentation and discussion with the Human Resources (HR) department. This session provided insight into the organisational management systems and workforce strategy underpinning Waaree's 10,000-employee operation. The HR function at Waaree is structured across four primary verticals:

4.1.1 Talent Acquisition

Waaree's recruitment pipeline operates through multiple channels: active sourcing from job portals (Naukri, LinkedIn), campus recruitment drives at engineering colleges and polytechnics (particularly relevant for diploma-level technician recruitment for the manufacturing floor), and employee referral programmes. The initial screening involves profile review, telephonic screening, and technical and HR interview rounds. Given the specialised nature of PV manufacturing, the company prioritises candidates with backgrounds in electrical engineering, electronics, materials science, and mechanical engineering for technical roles, while business roles are sourced from management institutes.

4.1.2 On-Boarding

New employees undergo a structured induction programme that includes an introduction to Waaree's history, vision, and values; briefing on HR policies (leave, benefits, code of conduct); Learning and Development (L&D) plan alignment; and CTC (Cost to Company) structure review. The induction is designed to ensure new hires are productive and culturally aligned as quickly as possible. For manufacturing floor employees, this induction is supplemented by line-specific technical training, safety certification, and a supervised probation period on the production line.

4.1.3 Operations — PMS and Technology

The HR operations function manages day-to-day workforce administration using SAP (Systems, Applications and Products) — a large-scale ERP (Enterprise Resource Planning) software platform used by Waaree for payroll, attendance, performance management, and workforce analytics. Where bespoke functionality is required beyond SAP's standard modules, third-party vendors are engaged to develop custom solutions.

Performance is managed through a structured Performance Management System (PMS) aligned to a Management by Objectives (MBO) framework, whereby individual KRAs (Key Result Areas) and KPIs (Key Performance Indicators) are cascaded from corporate strategic objectives through the Balanced Scorecard (BSC) framework. This ensures measurable alignment between individual performance targets and company-level goals. The PMS cycle typically includes mid-year reviews and annual appraisals, with outcomes linked to increment, promotion, and development decisions.

4.1.4 Payroll

The payroll function manages compensation processing for all 10,000 employees, including statutory deductions such as Provident Fund (PF) contributions (employer and employee, typically 12% of basic salary each), Employee State Insurance (ESI), professional tax, and income tax (TDS). Given the multi-state manufacturing footprint of Waaree, payroll compliance must adhere to labour laws across multiple jurisdictions.

4.2 Testing Laboratory

The testing laboratory is one of the most scientifically rich environments in the plant. It serves as the quality gatekeeping function, ensuring that modules leaving the factory meet or exceed the specifications claimed on their datasheets and the requirements of international certification standards. The laboratory was observed to operate across two main categories of testing: in-line production testing and type-approval/certification testing.

4.2.1 Sun Simulator — Flash Xenon Test

The centrepiece of the laboratory is the sun simulator — a large automated flash tester that replicates the standard solar test condition (STC) of 1,000 W/m² irradiance, AM1.5G spectrum, and 25°C cell temperature. A high-intensity xenon flash lamp provides the irradiation. Xenon is used because its emission spectrum closely matches the standard solar spectrum (AM1.5G) across the relevant wavelength range of 300–1,200 nm.

During the flash (duration ~1–10 ms), the module's IV curve is swept electronically at high speed — a voltage sweep from short-circuit to open-circuit is completed in the duration of the flash. The resulting IV curve is analysed by the system's software to extract all key electrical parameters: I_sc, V_oc, P_max, I_mp, V_mp, fill factor, and efficiency.

Based on their measured P_max values, modules are 'binned' (sorted) into power classes designated I_1, I_2, and I_3 — representing successively higher power output tiers within the same nominal wattage class. This binning ensures that modules in the same installation produce matched currents (critical for series-stringing in field installations), and allows premium pricing for higher-performance units. The plant's observed efficiency of approximately 25% for TOPCon modules is consistent with state-of-the-art commercial production.

> Module efficiency: $\eta = P_{max} / (G_{STC} \times A_{module})$ where $G_{STC} = 1000\ W/m^2$

4.2.2 Electroluminescence (EL) Test

Electroluminescence (EL) imaging is a powerful non-destructive quality inspection technique that provides a spatially resolved 'map' of the electrical quality of the solar cell and module. In EL testing, a forward-bias DC current is injected into the module (reversing the normal operating mode). The injected current causes radiative recombination of electron-hole pairs within the silicon, emitting near-infrared photons (wavelength ~1,150 nm for silicon) in proportion to the local minority carrier lifetime.

A cooled CCD (charge-coupled device) camera with sensitivity in the near-infrared range captures the emitted luminescence pattern. Defect-free, high-lifetime regions emit strongly and appear bright; regions with cracks, broken interconnections, poor soldering, or high recombination rates appear dark.

The EL test provides direct visualisation of: micro-cracks (finger-like dark lines), broken busbars (entire cell rows dark), delamination (irregular dark patches), and current mismatch between cell strings. It is performed after lamination (when physical repairs are no longer possible) to classify modules according to a three-tier quality grading system:

Grade	Designation	EL Characteristics	Disposition
A	Perfect	No cracks, uniform luminescence across all cells	Ship to customer; full warranty
B	Defect but Usable	Minor inactive area (<2% of cell area); no cross-cracks	Sold at discount; restricted warranty
R	Reject	Cracks affecting >2% of cell area; broken ribbons; delamination	Recycled or scrapped; not shipped

Table 4.1 — EL test quality grading system at Waaree.

4.2.3 Laminator Quality Test (VG100)

A dedicated quality check is performed on the laminator's output using the VG100 system, which applies both vacuum (to test hermetic sealing integrity) and controlled pressure to assess the lamination bond strength. Adequate lamination is essential for long-term module durability: delamination — the separation of the EVA encapsulant from either the glass or the cell surface — is a common field failure mode that leads to moisture ingress, corrosion of the metal contacts, and accelerated power degradation. The VG100 test verifies that the peel strength of the EVA-glass interface exceeds a minimum threshold throughout the module area.

4.2.4 IEC 61215:2016 Type Approval Testing

Beyond the in-line production tests, Waaree's modules must undergo IEC 61215:2016 (formerly IEC 61215:2005) type-approval certification — the internationally recognised standard for terrestrial PV module design qualification. This certification, conducted by accredited third-party laboratories (such as TÜV Rheinland or Bureau Veritas), subjects modules to a comprehensive suite of five test sequences designed to simulate 25 years of field exposure:

- Thermal cycling (200 cycles, −40°C to +85°C): evaluates resistance to fatigue failure from diurnal temperature swings.
- Damp heat (1000 hours at 85°C, 85% relative humidity): accelerated corrosion and delamination test.
- Humidity freeze (10 cycles, 85% RH to −40°C): evaluates moisture infiltration and freeze-thaw damage.
- Mechanical load (5,400 Pa static load, front and rear): simulates snow and wind loading.
- Hail impact (25 mm ice balls at 23 m/s): simulates hail damage.

Passing IEC 61215 is mandatory for module bankability — the property that makes a module acceptable as collateral for project finance. Without IEC 61215 certification, no commercial solar project developer or bank will accept Waaree's modules for large-scale installation.

4.2.5 QR Code and RFID Traceability

Throughout the manufacturing process — from cell production through module assembly to post-testing — each unit is tagged with a QR code and RFID (Radio-Frequency Identification) label. This enables complete cradle-to-grave traceability: every module can be traced back to its constituent cells, the specific production line and shift that manufactured it, all in-process test results, and the batch of raw materials used. This traceability system is critical for warranty management (enabling rapid identification and field replacement of modules from a defective production batch), regulatory compliance, and continuous quality improvement.

4.2.6 CLB Process Presentation

The Day 2 programme also included a technical presentation on the Cell-Level production Line (CLB) process, which elaborated on the semiconductor manufacturing steps observed in abbreviated form during the Day 1 line visit (and described in full detail in Section 5 of this report). The presentation covered the physics of p-n junction formation, the role of anti-reflection coatings, and the efficiency roadmap from PERC to TOPCon to heterojunction technology.

5. Day 3 Observations — Quality Control, Cell Manufacturing & Maintenance

5.1 Quality Control Department

The Quality Control (QC) department at Waaree operates as an independent function with authority to hold or reject production at any stage of the manufacturing process. Its remit spans incoming materials inspection, in-process statistical process control (SPC), final product testing, and customer complaint analysis. The QC department's activities are structured around three internationally recognised quality management frameworks:

5.1.1 ISO Quality and Environmental Management

Waaree holds ISO 9001:2015 (Quality Management System) certification, which provides the overarching framework for its quality management system. ISO 9001 mandates a process-based, risk-focused approach to quality, requiring documented procedures, measurable quality objectives, regular internal audits, management reviews, and a structured corrective action process. The certification is assessed by an accredited third-party certification body and renewed through annual surveillance audits and triennial re-certification audits.

In addition to ISO 9001, the facility maintains ISO 14001:2015 (Environmental Management System) certification, covering its management of environmental impacts — specifically soil contamination, water discharge, and air emissions — from the manufacturing processes. This is particularly relevant given the use of hazardous process gases in the CLB ($POCl_3$, BCl_3, HF) and the generation of silicon wafer cutting waste (silicon kerf slurry).

5.1.2 The 6S Workplace Organisation System

Throughout the production floor, the 6S Japanese workplace organisation methodology was visibly implemented. Derived from the Toyota Production System (TPS) and Lean Manufacturing philosophy, 6S provides a systematic framework for organising the shop floor to improve efficiency, reduce waste, and enhance safety. The six 'S' principles, as explained during the visit, are:

S	Japanese Term	Meaning	Observed Application at Waaree
1 — Sort	Seiri	Remove all unnecessary items from the workplace	Only tools and materials required for current production present at each station; red-tag system for questionable items
2 — Set in Order	Seiton	A place for everything; everything in its place	Shadow boards for tools; floor markings for equipment positions; labelled storage locations
3 — Shine	Seiso	Clean workspaces; identify and eliminate sources of contamination	Daily cleaning schedules; equipment maintained in clean condition; immediate cleanup of spills
4 — Standardise	Seiketsu	Create visual controls and standard conditions	SOPs posted at each workstation; colour-coded pipework; visual management displays
5 — Sustain	Shitsuke	Maintain and improve the first four S's	Regular 6S audits; management walkthrough schedule; employee accountability

S	Japanese Term	Meaning	Observed Application at Waaree
6 — Safety	Anzen	Integrate safety into all workplace organisation activities	Safety markings on floor; PPE storage at entry points; emergency stop buttons accessible

Table 5.1 — 6S principles and their observed application at Waaree Chikhli.

The 6S system, when properly sustained, delivers measurable manufacturing outcomes: increased line efficiency (reduced time searching for tools or materials), reduced defect rates (cleaner environments produce fewer contamination-related defects), reduced equipment downtime (clean, well-maintained equipment fails less frequently), and reduced workplace accidents. The use of Standard Operating Procedures (SOPs) at every workstation ensures that process parameters are consistently applied regardless of which operator is present — a critical requirement for achieving the repeatability demanded by IEC 61215 certification.

5.1.3 Core Values and Manpower Agency Framework

The QC department presentation also touched on Waaree's core organisational values — Safety, Integrity, and Customer Care — which form the ethical backbone of its quality culture. The workforce includes employees from four manpower agencies, a common structure in Indian manufacturing that allows operational flexibility while maintaining a core permanent workforce. The integration of contract workers into the quality culture (ensuring they follow SOPs and 6S practices equally with permanent employees) is an ongoing HR-QC interface challenge that Waaree manages through standardised induction and regular retraining.

5.2 Cell-Level Production Line (CLB) — Detailed Process

The Cell-Level production Line (CLB) manufactures the solar cells from incoming silicon wafers. This is the most technically complex and scientifically rich part of the facility, involving a sequence of high-temperature diffusion furnaces, chemical etch baths, plasma deposition systems, and precision screen printing machines. The following process sequence was explained in detail and is annotated with the relevant semiconductor physics.

5.2.1 From Silicon to Wafer

The upstream value chain preceding the CLB, though not fully visible during the visit, was explained in briefing sessions. The journey from raw silicon to wafer involves:

- Metallurgical-grade silicon (MG-Si, ~98% pure) is produced by carbothermic reduction of quartz (SiO_2) with carbon in an electric arc furnace.
- Purification to solar-grade polysilicon (>99.9999% pure, 6N) via the Siemens process (trichlorosilane decomposition) or the Fluidised Bed Reactor (FBR) route.
- Crystal growth: polysilicon is melted and directionally solidified into monocrystalline silicon ingots (boules) via the Czochralski (CZ) process, where a seed crystal is slowly withdrawn from a rotating silicon melt in a precise temperature gradient.
- Wire sawing: ingots are sliced into thin wafers (approximately 160–180 µm thick) using diamond wire saws. This cutting produces silicon kerf (waste powder), currently around 40% of the ingot mass — recovery and recycling of kerf is an active area of research.

5.2.2 PERC Cell Manufacturing Process

The PERC (Passivated Emitter and Rear Cell) architecture uses a p-type silicon base with specific passivation of the rear surface to reduce carrier recombination. The full process sequence as described during the CLB presentation is:

1	**Raw Wafer Incoming**	P-type CZ silicon wafers (~180 µm, 182×182 mm). Initial resistivity ~1–3 Ω·cm. Incoming quality inspection: thickness, TTV (total thickness variation), bow, crack detection.
	↓	
2	**Texturing**	Alkaline KOH etch creates random pyramid microstructure on wafer surface. Pyramids ~3–10 µm height. Purpose: reduces front surface reflectance from ~35% (flat) to <5% (textured) by trapping light through multiple reflection events. $Si + 2KOH + H_2O \rightarrow K_2SiO_3 + 2H_2\uparrow$
	↓	
3	**$POCl_3$ Diffusion (Emitter)**	Wafers loaded into tube diffusion furnace at ~800–850°C in $POCl_3$ + O_2 atmosphere. Phosphorus diffuses into surface ~300–500 nm, forming n+ emitter layer (sheet resistance ~60–90 Ω/sq). Creates p-n junction ~0.3–0.5 µm below front surface.
	↓	
4	**Selective Emitter Formation**	Laser doping or etch-back creates heavily doped n++ regions (low resistance) only under the metal contact fingers. Lightly doped regions (n+) elsewhere minimise recombination under glass. Improves V_oc and FF by reducing contact resistance.
	↓	
5	**Pre-Annealing / PSG Removal**	Phosphosilicate glass (PSG) layer formed during $POCl_3$ diffusion is etched away in dilute HF solution. Pre-anneal at ~700°C begins oxide passivation quality improvement. Prepares surface for subsequent PECVD deposition.
	↓	
6	**LCO (Laser Contact Opening)**	Laser ablates precise openings in the rear dielectric passivation layer at contact points. Allows rear metal to contact silicon without disrupting the passivating oxide across the rest of the rear surface.
	↓	
7	**Front PECVD (SiN_x:H ARC)**	PECVD (Plasma-Enhanced CVD) deposits silicon nitride (SiN_x:H) anti-reflection coating on front surface. $SiH_4 + NH_3 \rightarrow SiN_x$:H + byproducts (in RF plasma at ~400°C) Thickness ~75–80 nm; refractive index ~2.0. Minimises reflection at λ~600 nm. Hydrogen passivates bulk defects during deposition.
	↓	
8	**Rear PECVD (Al_2O_3 + SiN_x)**	Alumina (Al_2O_3) deposited by PECVD or ALD on rear surface. Provides negative fixed charge that repels minority carrier holes — field-effect passivation. Capped with SiN_x layer for additional passivation and protection. This dual-layer rear passivation is the defining feature of PERC architecture.
	↓	
9	**Post-Annealing (FGA)**	Forming gas anneal (H_2/N_2 atmosphere at ~350–450°C). Drives hydrogen from SiN_x layer into Si bulk to passivate dangling bonds at grain boundaries and at Si/SiO_2 interface. Significantly reduces bulk and interface recombination; improves V_oc.
	↓	

10	**Screen Printing — Metallisation**	Front: silver paste screen-printed as fine fingers (30–40 µm width) and busbars. Rear: aluminium paste printed on rear contact areas; silver pads for ribbon connection. Squeegee presses paste through stencil mesh; alignment ±10 µm accuracy required.
	↓	
11	**Co-Firing**	Conveyer belt furnace with peak temperature ~800–850°C. Front Ag: burns through SiN_x ARC and forms ohmic contact to n+ emitter. Rear Al: forms Al-doped p+ BSF (back surface field) layer — creates local p+p junction that repels minority electrons. All contacts formed in a single rapid thermal step (~30s peak).
	↓	
12	**Cell Testing & Binning**	Flash IV test on every cell under STC. Current measured to ±0.1 mA accuracy. Cells sorted into I_1, I_2, I_3 bins by I_sc and V_oc. EL imaging for crack and defect detection. Quality grade A/B/R assigned.

Figure 5.1 — PERC solar cell manufacturing process sequence at Waaree CLB.

5.2.3 TOPCon Cell Manufacturing — Key Differences

TOPCon (Tunnel Oxide Passivated Contact) technology represents the next generation beyond PERC and is increasingly the focus of Waaree's cell manufacturing capacity. The architecture uses an n-type silicon base (phosphorus-doped) with a novel rear contact structure consisting of a thin (~1.5 nm) silicon oxide tunnel layer (SiO_x) topped by a heavily doped polysilicon (poly-Si) layer.

The tunnel oxide allows majority carriers (electrons in n-type) to pass through by quantum mechanical tunnelling, while blocking minority carriers (holes) from reaching the rear metal contact — dramatically reducing rear surface recombination. This gives TOPCon cells their superior open-circuit voltage (V_oc > 710 mV vs. ~680 mV for PERC) and consequently higher efficiency (24–26% vs. 22–24%).

The key manufacturing differences from PERC are:

- BCl_3 diffusion (instead of $POCl_3$) for boron-doped p+ emitter on front surface.
- LPCVD (Low-Pressure CVD) or PECVD deposition of the ~1.5 nm tunnel oxide and n+ polysilicon rear stack — a technically demanding step requiring tight thickness control.
- More bus bars (typically 9–12 busbars in TOPCon vs. 5–6 in PERC) to reduce resistive path length from the narrow cell fingers to the ribbon — enabled by the lower contact recombination of the passivated rear.

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TOPCon V_oc advantage: ΔV_oc ≈ (nkT/q) × ln(J_0,PERC /
                       J_0,TOPCon) ≈ +30 mV
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Note: *The efficiency hierarchy observed at Waaree — PERC < TOPCon < Heterojunction (HJT) — reflects the progressive improvement in surface and interface passivation quality. HJT cells, which use amorphous silicon (a-Si:H) layers deposited at low temperature to passivate both surfaces, achieve V_oc values approaching 750 mV and efficiencies of 26–27%, but require expensive ITO (indium tin oxide) front contacts and low-temperature silver paste, significantly increasing manufacturing cost.*

5.3 Maintenance Department

The Maintenance department ensures operational continuity of the highly automated, capital-intensive manufacturing equipment. In a facility where a single laminator may cost

several crore rupees and a PECVD system tens of crore, unplanned downtime has a direct and severe impact on revenue and customer delivery commitments.

The maintenance function at Waaree operates on a Total Productive Maintenance (TPM) philosophy, which shares conceptual roots with the 6S system in emphasising proactive, systematic equipment care rather than reactive breakdown maintenance. Key maintenance activities observed and described include:

- Preventive Maintenance (PM) schedules: every piece of equipment has a documented PM schedule specifying daily, weekly, monthly, and annual maintenance tasks. For the laminator, this includes daily cleaning of the membrane and heating plates, weekly EVA residue removal, and monthly calibration of temperature sensors and pressure gauges.
- Predictive Maintenance (PdM): vibration monitoring, thermal imaging (thermography) of electrical panels, and oil analysis for gearboxes in the conveyer drives are used to detect developing faults before they cause failure — enabling planned maintenance during scheduled downtime windows rather than emergency breakdown repairs.
- Spare parts management: critical spare parts (PECVD electrodes, stringer heating elements, laminator membranes) are stocked onsite to minimise mean time to repair (MTTR). The trade-off between the cost of carrying spare parts inventory and the cost of downtime is managed using an ABC analysis framework.
- Equipment OEE (Overall Equipment Effectiveness): each production machine's OEE — the product of availability, performance, and quality rate — is tracked on visible production dashboards, providing the maintenance team with real-time visibility of equipment health and production performance.

6. Complete Manufacturing Process Flow Summary

The following diagram summarises the complete end-to-end manufacturing process flow from silicon raw material to dispatched module, integrating both the CLB (cell manufacturing) and MLB (module assembly) stages:

1	**SILICON RAW MATERIAL**	Metallurgical grade Si → solar-grade polysilicon (Siemens/FBR process) → CZ crystal growth → ingot → wire sawing → wafer (182×182 mm, ~180 µm)
	↓	
2	**WAFER INCOMING INSPECTION**	Thickness, resistivity, bow, TTV, visual defect check. P-type for PERC; N-type for TOPCon.
	↓	
3	**TEXTURING**	KOH wet etch → random pyramid microstructure. Reflectance reduced from ~35% to <5%.
	↓	
4	**DIFFUSION / DOPING**	$POCl_3$ (PERC p-type) or BCl_3 (TOPCon n-type) tube furnace diffusion. Forms p-n junction.
	↓	
5	**SELECTIVE EMITTER + ETCH**	Laser doping of contact regions. PSG removal in HF. Prepares optimised emitter profile.
	↓	
6	**PECVD / PASSIVATION LAYERS**	SiN_x:H ARC on front; Al_2O_3 + SiN_x on rear (PERC). Or tunnel oxide + poly-Si (TOPCon).
	↓	

7	ANNEALING	Forming gas anneal at 350–450°C. Hydrogen passivation of bulk and interface defects.
	↓	
8	SCREEN PRINTING	Silver paste (front fingers + busbars) and aluminium paste (rear) via screen printing.
	↓	
9	CO-FIRING	Belt furnace ~850°C peak. Contact formation; Al-BSF for PERC. All contacts in single step.
	↓	
10	CELL TEST & EL INSPECTION	Flash IV test (STC). EL imaging. Binning $I_1/I_2/I_3$. Grade A/B/R assignment.
	↓	
11	LASER CUTTING (MLB START)	Full cells laser-cut to half-cells (182×91 mm). 1/2 of each pair flipped for current uniformity.
	↓	
12	STRINGER	Copper ribbons (coated, 0.4 mm) fluxed, aligned and soldered to busbars. Strings of 12 half-cells. I-type/Z-type/U-type bus bars.
	↓	
13	LAY-UP / SANDWICH ASSEMBLY	Back glass → EVA → cell strings → EVA → front glass (anti-reflective, >95% transmittance). Robotic arms + suction cups.
	↓	
14	LAMINATION	Vacuum press at 120°C then 150°C. EVA crosslinks to transparent solid. Hermetic seal formed.
	↓	
15	FRAMING + JUNCTION BOX	Aluminium frame bonded with silicone sealant (4 hr cure). J-box with bypass diodes installed. MC4 cables terminated.
	↓	
16	SUN SIMULATOR TEST	Flash xenon lamp IV sweep. P_max measured. Binned $I_1/I_2/I_3$ by voltage. ~25% efficiency.
	↓	
17	EL + FINAL QC	EL imaging of completed module. Visual inspection. IEC 61215 compliance verification.
	↓	
18	QR/RFID TAGGING & DISPATCH	Module assigned unique serial number with full production traceability data. Packaging, palletisation, and dispatch to customer.

Figure 6.1 — Complete Waaree solar PV manufacturing process flow (CLB + MLB integrated).

7. Cell Technology Comparison: PERC vs TOPCon vs Heterojunction

A recurring theme throughout the visit was the technology transition under way in the PV manufacturing industry from PERC to TOPCon. The following table provides a rigorous comparative analysis of the three main cell technologies discussed:

Parameter	PERC	TOPCon	Heterojunction (HJT)
Base type	P-type CZ silicon	N-type CZ silicon	N-type CZ silicon
Emitter formation	$POCl_3$ diffusion	BCl_3 diffusion	a-Si:H PECVD (low T)
Rear passivation	Al_2O_3/SiN_x stack	Tunnel oxide + n+ poly-Si	a-Si:H + ITO
Front contact	Ag screen print	Ag screen print	ITO + low-T Ag paste
Typical V_oc	~680 mV	~710–720 mV	~740–750 mV
Typical efficiency	22–24%	24–26%	25–27%
No. of busbars	5–6	9–12	varied (busbarless possible)
Key advantage	Mature; low cost	High V_oc; cost competitive	Highest V_oc; bifacial
Key limitation	LID degradation; BSF recombination	LPCVD cost; boron contamination	ITO cost; low-T process sensitivity
Bifacial capable	Yes (limited)	Yes	Yes (excellent)
Manufacturing temp.	Up to 850°C	Up to 850°C	<200°C (a-Si:H steps)

Table 7.1 — Comparative analysis of PERC, TOPCon, and Heterojunction solar cell technologies.

8. Business Analysis: Strategic Positioning and Competitive Landscape

8.1 Vertical Integration as Competitive Advantage

Waaree's most distinctive strategic asset is its degree of vertical integration. Unlike most Indian PV manufacturers who assemble modules from imported cells (primarily from China), Waaree manufactures cells from wafers at Chikhli, and has invested in wafer and electrolyser manufacturing capabilities. This vertical integration provides several competitive advantages:

- Cost control: internal cell supply eliminates cell import margin and reduces exposure to international supply chain disruptions (as experienced during COVID-19 and the India-China trade tensions of 2020).
- Quality control: complete control of the manufacturing process from wafer to module enables rapid corrective action when quality issues arise, rather than depending on supplier response cycles.
- Technology leadership: in-house cell manufacturing allows Waaree to adopt new cell technologies (PERC to TOPCon) on its own timeline rather than being dependent on cell supplier technology roadmaps.
- Government policy alignment: India's PLI (Production-Linked Incentive) scheme specifically rewards integrated domestic manufacturing, providing additional financial benefit to companies like Waaree that manufacture locally across the value chain.

8.2 Green Hydrogen — Strategic Diversification

Waaree's investment in electrolyser manufacturing represents a forward-looking strategic diversification into the nascent green hydrogen economy. Green hydrogen is produced by water electrolysis powered by renewable electricity — a process that uses PEM (Proton Exchange Membrane) or alkaline electrolysers. The synergy with Waaree's core solar business is clear: as the cost of solar electricity continues to decline, green hydrogen becomes increasingly cost-competitive with 'grey' hydrogen (produced from natural gas), making electrolyser manufacturing a strategically aligned adjacency.

The Indian government's National Green Hydrogen Mission (NGHM), launched in 2023, has set a target of 5 million tonnes of green hydrogen production per year by 2030, representing an enormous potential market for electrolyser manufacturers like Waaree.

8.3 Market Position and Financial Metrics

Metric	Value	Context
Revenue (FY2024–25 approx.)	₹7,500 crore	Significant YoY growth; driven by domestic and export module demand
Market Capitalisation	₹25,000 crore	Reflects premium valuation for integrated manufacturer + growth expectations
Total Manufacturing Capacity	23.75 GW	Largest in India; competitive with mid-tier global manufacturers
Chikhli Cell Capacity	5.4 GW	Largest single-site cell factory in India
Employees	~10,000	80% operational; reflects labour-intensive manufacturing base
Manufacturing Facilities	9 sites	Geographic diversification across India
Export Markets	USA, Europe, Middle East	Premium markets; higher ASP than domestic; dollar revenue hedge

Table 8.1 — Waaree Energies key business metrics and context.

9. Critical Observations and Reflections

9.1 Strengths Observed

- Scale and automation: the level of automation in both MLB and CLB lines — with robotic handling, automated optical inspection, and SPC-linked process control — is consistent with Tier-1 global standards and minimises human-introduced variability.
- Traceability infrastructure: the QR/RFID end-to-end traceability system represents best practice in manufacturing quality management and significantly reduces warranty risk.
- Technology migration discipline: the structured approach to transitioning from PERC to TOPCon — running both technologies in parallel while scaling the higher-efficiency product — demonstrates mature manufacturing strategy.
- 6S implementation: the visible, sustained implementation of 6S on the production floor indicates a genuine quality culture rather than a superficial compliance exercise.

9.2 Limitations and Challenges Observed

- Silicon kerf waste: the wire sawing process generates approximately 40% of the ingot mass as silicon slurry waste. While recycling technologies exist, their economic viability at scale remains limited. This represents both an environmental challenge and a material cost inefficiency.
- Silver consumption: screen-printed silver contacts represent the highest per-cell material cost (silver price ~₹75,000/kg). The industry trend toward silver reduction (thinner fingers, busbarless designs, copper plating alternatives) is a critical cost-reduction lever that requires continuous process development investment.
- Dependence on imported materials: despite domestic cell manufacturing, key raw materials — polysilicon, silver paste, EVA film, anti-reflection coated glass — are substantially imported, maintaining supply chain and currency exposure.
- Water consumption in cell manufacturing: the HF etch and KOH texturing processes require significant volumes of ultra-pure water, creating both operational cost and environmental water stewardship obligations.

9.3 Future Technology Trajectory

Based on the technical presentations and observations during the visit, the following technology developments appear to be on Waaree's near-term roadmap:

- Full migration to TOPCon: given TOPCon's ~2 percentage point efficiency advantage over PERC, new capacity additions are expected to be exclusively TOPCon, with PERC lines eventually repurposed or decommissioned.
- Heterojunction (HJT) evaluation: HJT's superior bifacial performance and high V_oc make it the leading candidate for Waaree's next-generation product, though the low-temperature deposition requirement and ITO cost remain barriers to rapid adoption.
- Tandem cells: perovskite-silicon tandem cells, which stack a perovskite top cell over a silicon bottom cell to capture a broader range of the solar spectrum, offer theoretical efficiencies above 40% and are the subject of intensive global R&D. Commercial tandem modules may enter the market within 5–7 years.
- Thinner wafers: reducing wafer thickness from ~180 µm toward 100–130 µm would reduce silicon consumption per cell by ~30%, significantly reducing cost per watt — contingent on development of handling and processing equipment for thinner, more fragile wafers.

10. Conclusion

This three-day observation visit to Waaree Energies' Chikhli manufacturing campus provided an exceptionally detailed view of the scientific, engineering, and organisational complexity underlying the production of a commercial solar photovoltaic module. From the quantum mechanical principles of silicon semiconductor physics — band gap theory, p-n junction formation, and the photovoltaic effect — through the precision materials processes of diffusion, PECVD, and screen printing, to the automated assembly, lamination, and quality certification of finished modules, the visit revealed a tightly integrated, scientifically sophisticated manufacturing system operating at extraordinary scale.

The company's trajectory — from a precision instrument manufacturer in 1989 to India's largest integrated solar manufacturer with 23.75 GW capacity and green hydrogen ambitions in 2025 — reflects a combination of strategic foresight, patient capital investment, and disciplined operational execution. The quality management systems observed (ISO 9001, ISO 14001, IEC 61215, 6S, SOP, QR/RFID traceability) represent a mature, internationally benchmarked approach to manufacturing quality that is a prerequisite for participation in the global bankable solar market.

The technology transition from PERC to TOPCon, visible in the coexistence of both cell architectures on the production floor, reflects Waaree's positioning at the frontier of commercial PV technology — upgrading its product without disrupting its manufacturing base. The strategic investments in green hydrogen electrolysers and end-to-end EPC services represent a coherent long-term vision of Waaree as a comprehensive clean energy solutions company rather than a commodity module manufacturer.

For any student of applied science, engineering, or business strategy, a visit to Waaree Chikhli offers a rare opportunity to observe fundamental scientific principles — semiconductor physics, electrochemistry, materials science, optical engineering — operating simultaneously at industrial scale in service of one of the defining technological transitions of our era: the decarbonisation of the global energy system through photovoltaic power generation.

End of Report

WAAREE®
One with the Sun

Date: - 07th May, 2026

To Whomsoever It May Concern

I am pleased to recommend **Aarav A Jain**, Grade 11 of Step-by-Step School, Noida, who undertook an in-depth academic engagement involving a field study of solar photovoltaic manufacturing at Waaree Energies Ltd.

During the engagement from 23rd to 25th March 2026, Aarav gained exposure to industrial processes, quality control, and organizational practices, including ISO standards and 6S methodology. He learned semiconductor basics and the end-to-end production flow—from cell fabrication and module assembly to quality testing and high-volume manufacturing. He also showed strong interest in advanced solar technologies such as PERC and TOPCon. Throughout, he demonstrated intellectual curiosity and the ability to connect classroom concepts with real-world industrial applications.

His ability to grasp knowledge of integrated principles of physics and materials science, was commendable. Aarav's ability to synthesize complex, interdisciplinary concepts into a structured and analytical report was especially impressive. His engagement was clearly guided by a genuine commitment to environmental sustainability, which led him to explore solar energy as a practical and scalable solution to environmental problems.

Beyond his technical skills, Aarav showed maturity and professionalism throughout the engagement. He remained disciplined and eager to learn, asked thoughtful questions, contributed to cross-department discussions, and clearly communicated his observations and conclusions.

Aarav is a motivated and intellectually engaged person who brings both curiosity and rigor to his work. I believe this industrial exposure will be helpful in shaping his future career goals. I am confident that he will excel in academically challenging environments and contribute meaningfully to any institution he joins.

Waaree Energies Limited

WAAREE ENERGIES LIMITED

Vishal Upadhyay
Vice President- Human Resource